program 5

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5. Implement an R program to create a histogram illustarting the distribution of a continuous variable with overlays of density curves for each group, using ggplot2

step 1: Load necessary library

```
library(ggplot2)
```

step 2: Explore the inbuilt dataset

```
str(iris)
```

```
'data.frame':   150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

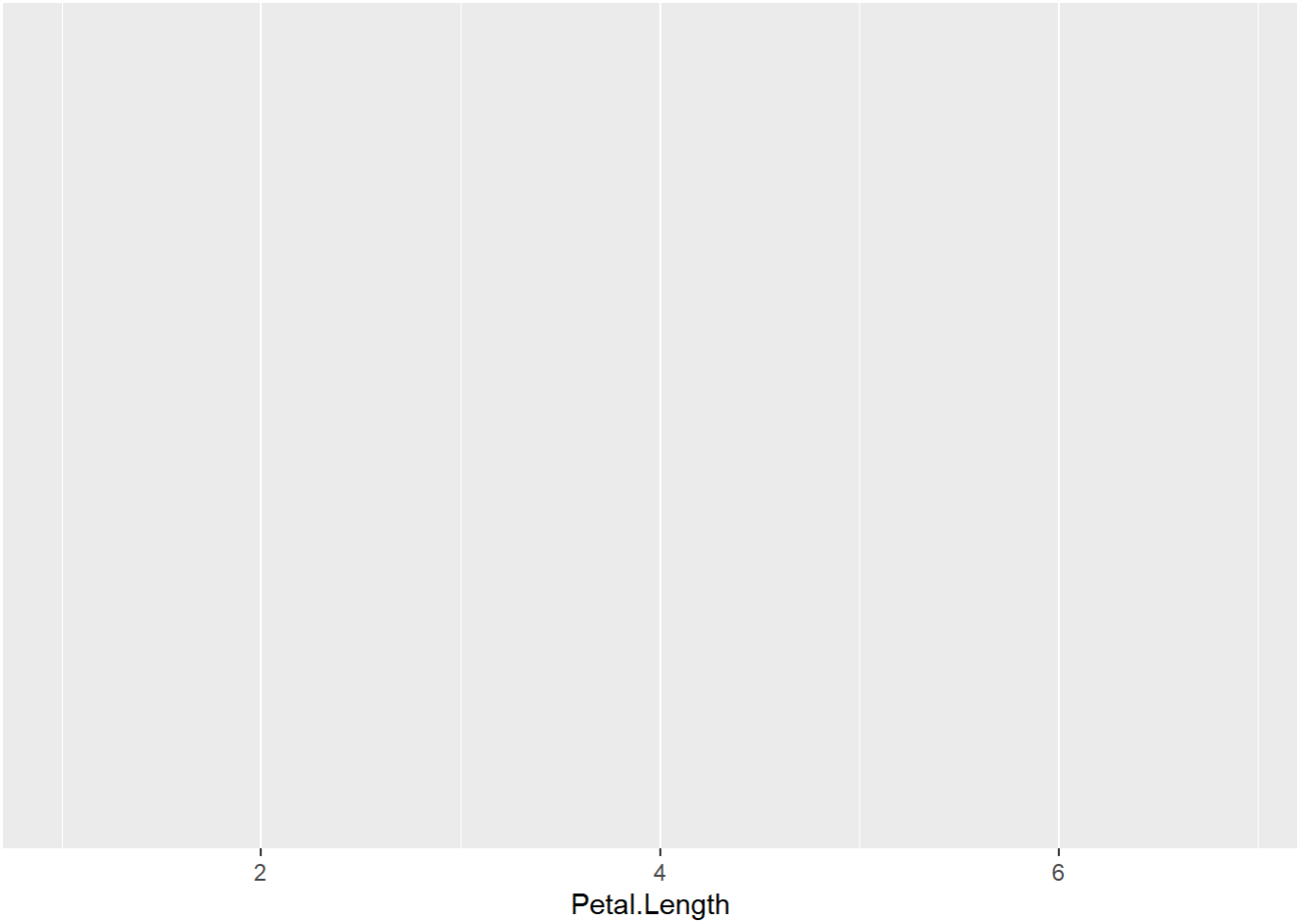
```
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

step 3:Create Hisstogram with Group-wise Density Curves

step 3.1: Initialize the ggplot with aesthetic mapping

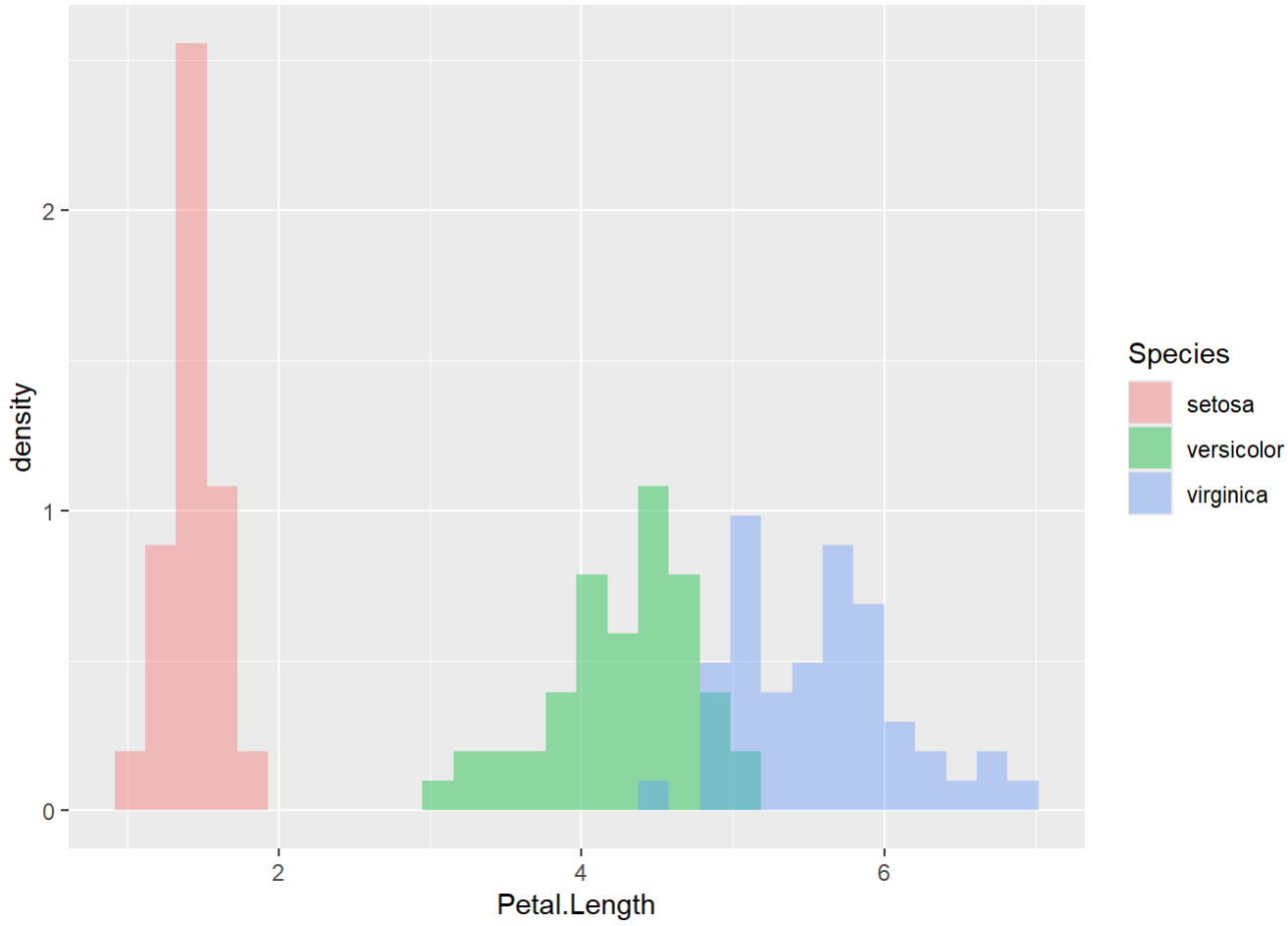
```
p <- ggplot(data = iris,aes(x=Petal.Length,fill = Species))
p
```



step 3.2: Add histogram layer

```
p <- p + geom_histogram(aes(y = ..density..),
                        alpha = 0.4,
                        position = "identity",
                        bins = 30)
p
```

```
Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
ℹ Please use `after_stat(density)` instead.
```

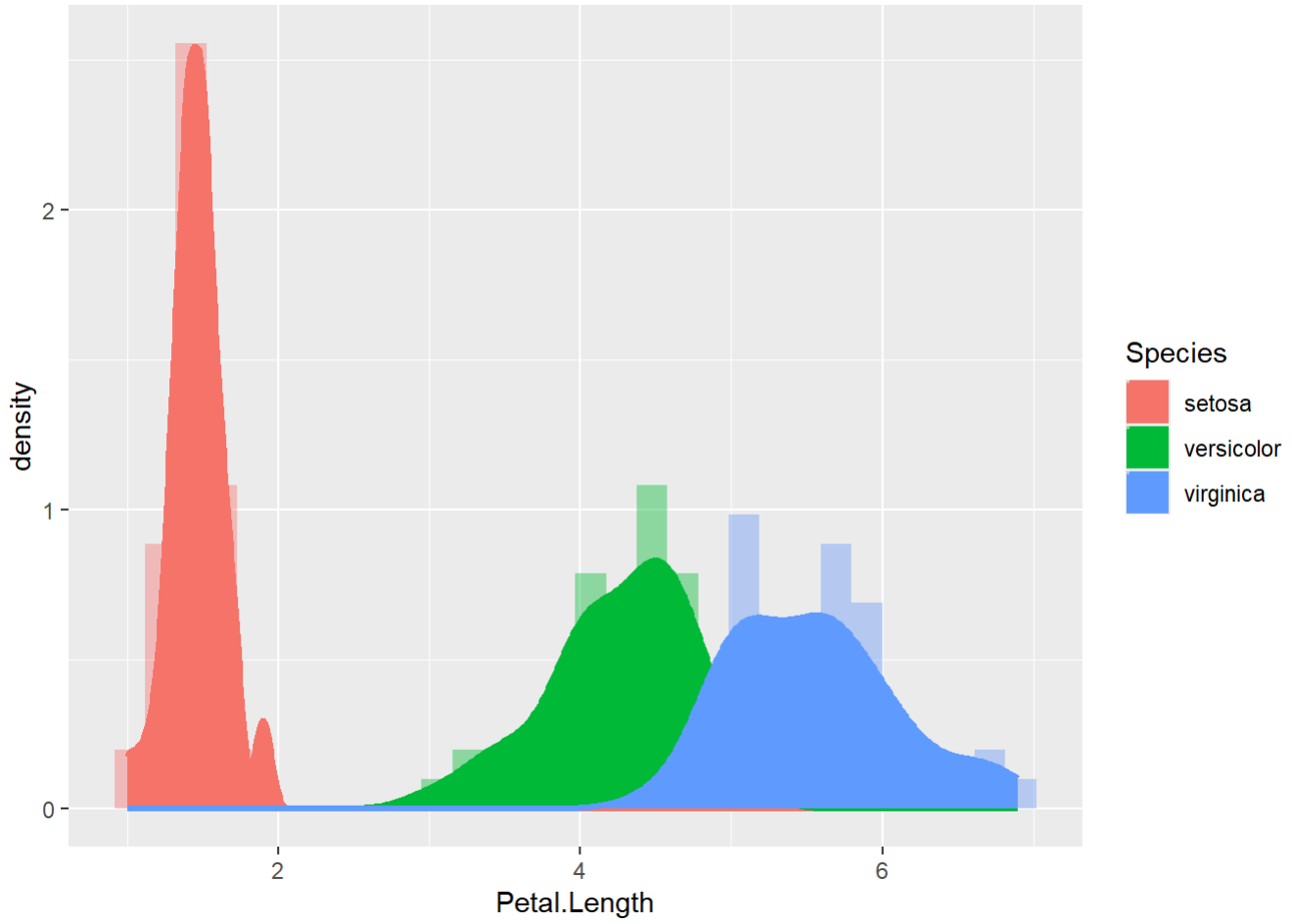


step 3.3: Add Density Curve layer

```
p <- p +
  geom_density(aes(color = Species),
               size = 1.2)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.
```

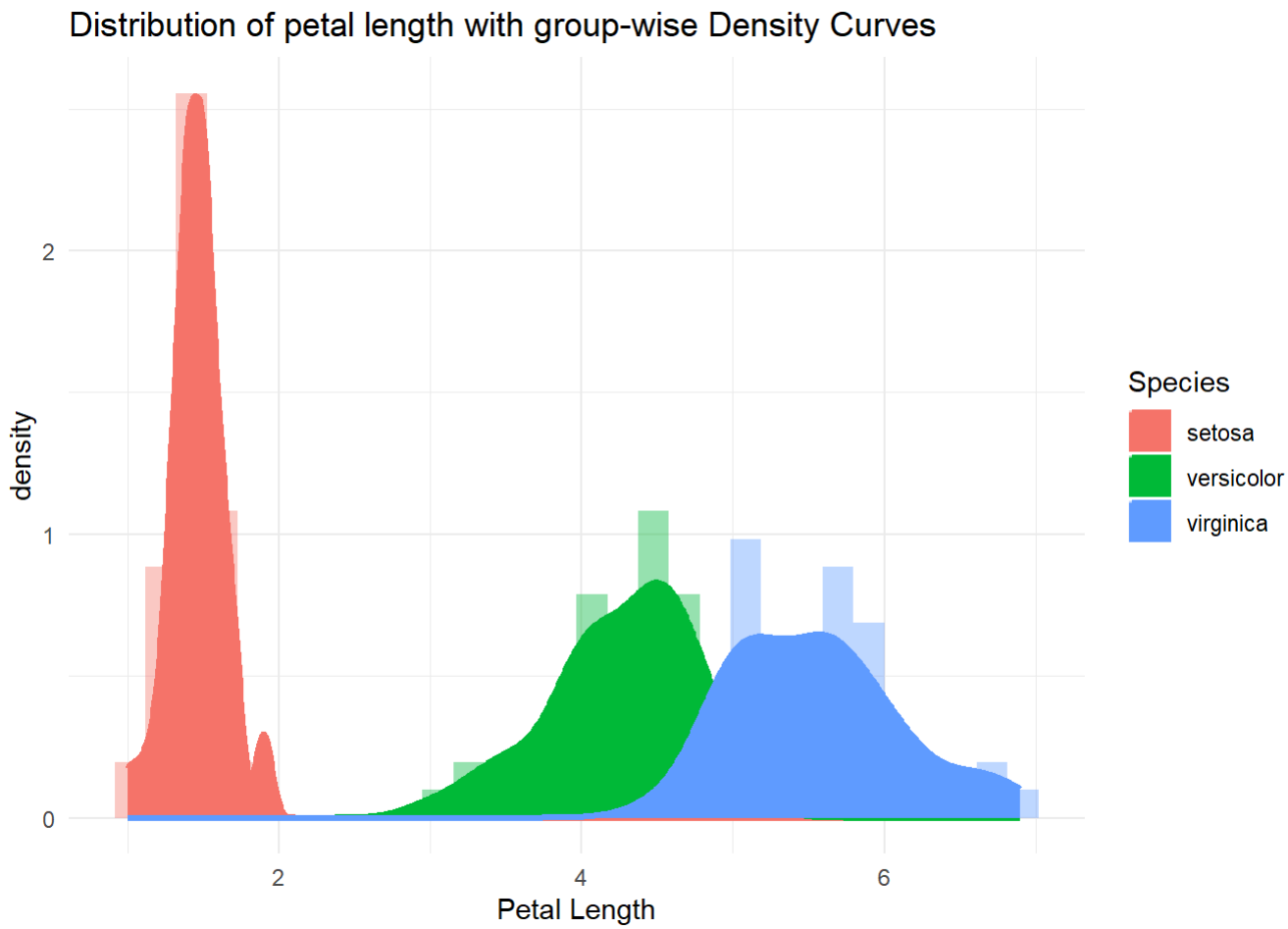
```
p
```



step 3.4:

```
p <- p + labs(
  title = "Distribution of petal length with group-wise Density Curves",
  x = "Petal Length",
  y = "density")+

theme_minimal()
p
```


Distribution of petal length with group-wise Density Curves
density
0
1
2
2
4
6
Petal Length
Species
setosa
versicolor
virginica